

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412877
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Hamano; Shinya et al.

Circuit assembly and electrical junction box

Abstract

A circuit assembly includes a plurality of electronic components; a plurality of control terminals electrically connected to the plurality of electronic components; and a holding member having insulating properties and holds the plurality of control terminals, wherein the control terminals serve as insert articles that form one piece with the holding member that is made of a resin, the control terminals each include: in a partial portion of the control terminal, an exposed face that is exposed from the holding member; a contact face that is located opposite to the exposed face and is in contact with the holding member; a first side face located between the exposed face and the contact face; and a second side face located between the exposed face and the contact face, and a first space formed lateral to the first side face, and a second space formed lateral to the second side face.

Inventors: Hamano; Shinya (Yokkaichi, JP), Sasaki; Yoshikazu (Yokkaichi, JP), Ikeda; Jun (Yokkaichi, JP)

Applicant: Sumitomo Wiring Systems, Ltd. (Yokkaichi, JP)

Family ID: 85477181

Assignee: Sumitomo Wiring Systems, Ltd. (Yokkaichi, JP)

Appl. No.: 17/934522

Filed: September 22, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230103105 A1	Mar. 30, 2023

Foreign Application Priority Data

JP	2021-155762	Sep. 24, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: **H01L25/16** (20230101); **H01L21/48** (20060101); **H01L23/13** (20060101); **H01L23/498** (20060101); **H01L25/065** (20230101)

U.S. Cl.:

CPC **H01L25/162** (20130101); **H01L21/4846** (20130101); **H01L23/13** (20130101); **H01L23/49838** (20130101); **H01L25/0655** (20130101);

Field of Classification Search

CPC: H01L (25/162)

USPC: 174/520

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6522528	12/2002	Yamane	361/601	B60R 16/0238
2002/0141143	12/2001	Yamane	N/A	N/A
2019/0164878	12/2018	Hattori et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112016006928	12/2018	DE	N/A
H10256291	12/1997	JP	N/A

OTHER PUBLICATIONS

German Patent and Trademark Office, Office Action dated Jul. 16, 2025, Application No. 10 2022 122 696.4. cited by applicant

Primary Examiner: Thompson; Timothy J

Assistant Examiner: Maiga; Sidi M

Attorney, Agent or Firm: Honigman LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority of Japanese Patent Application No. JP 2021-155762 filed on Sep. 24, 2021, the contents of which are incorporated herein.

TECHNICAL FIELD

(2) The present disclosure relates to a circuit assembly and an electrical junction box.

BACKGROUND

(3) JP 2019-96769A discloses a circuit assembly that includes a control board, a plurality of

switching elements that operate based on signals from the control board, and busbars constituting a power circuit. The switching elements are connected to the busbars using solder.

(4) JP 2019-96769A is an example of related art.

(5) The circuit assembly further includes, in addition to the busbar, a plurality of control terminals as constituent components of the power circuit. The control terminals are electrically connected to electronic components such as switching elements, and signals from the control board are transmitted to the electronic components via the control terminals.

(6) The circuit assembly further includes a holding member that has insulating properties and holds the control terminals. The holding member is made of a thermoplastic resin, and is molded by injection molding. The molding of the holding member may be performed in a manner such that the control terminals are placed as insert articles in a die for injection molding. FIG. 9 is a cross-sectional view showing part of the die 90 for injection molding, and a control terminal 91.

(7) By bringing part of the control terminal 91 into surface-contact with the die 90, the control terminal 91 can be positioned with respect to the die 90. However, during the injection molding, due to an injection pressure of a molten resin 99, the resin 99 may partially flow into the space between the die 90 and the control terminal 91, causing a risk in which as shown in FIG. 9, the control terminal 91 is inclined or deformed inside the die 90. If such a situation occurs, the control terminal 91 may come into contact with another conductor 92 such as a busbar that forms one piece with the holding member 95, and a short-circuit may occur.

(8) Therefore, it is an object of the present disclosure to provide a circuit assembly that can suppress a movement or deformation of a control terminal when a resin holding member is molded.

SUMMARY

(9) According to an aspect of the present disclosure, a circuit assembly includes: a plurality of electronic components; a plurality of control terminals electrically connected respectively to the plurality of electronic components; and a holding member that has insulating properties and holds the plurality of control terminals, wherein the control terminals serve as insert articles that form one piece with the holding member that is made of a resin, each of the control terminals includes: in a partial portion of the control terminal, an exposed face exposed from the holding member; a contact face that is located opposite to the exposed face and is in contact with the holding member; a first side face located between the exposed face and the contact face; and a second side face located between the exposed face and the contact face, and a first space is formed lateral to the first side face, and a second space is formed lateral to the second side face.

(10) According to an aspect of the present disclosure, an electrical junction box includes the above-described circuit assembly, and a cover that covers the circuit assembly.

Advantageous Effects

(11) According to the present disclosure, it is possible to suppress a movement or deformation of a control terminal due to an injection pressure of a molten resin when a resin holding member is molded.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating an electrical junction box including a circuit assembly according to an embodiment.

(2) FIG. 2 is a perspective view of the circuit assembly when viewed from the side opposite to that in FIG. 1.

(3) FIG. 3 is an exploded view of the circuit assembly from which a control board is detached.

(4) FIG. 4 is a cross-sectional view illustrating a control terminals and part of a holding member.

(5) FIG. 5 is a perspective view illustrating control terminals and part of the holding member.

(6) FIG. 6 is a cross-sectional view taken along a line V in FIG. 5.

(7) FIG. 7 is a cross-sectional view illustrating part of a die for molding the holding member that is made of a resin.

(8) FIG. 8 is a cross-sectional view illustrating an aspect in which the holding member is molded with the control terminal serving as an insert article.

(9) FIG. 9 is a cross-sectional view illustrating part of a die for injection molding, and a control terminal.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(10) Hereinafter, overviews of an embodiment of the present disclosure will be listed and described.

(11) First Aspect

(12) According to the present embodiment, a circuit assembly includes: a plurality of electronic components; a plurality of control terminals electrically connected respectively to the plurality of electronic components; and a holding member that has insulating properties and holds the plurality of control terminals, wherein the control terminals serve as insert articles that form one piece with the holding member that is made of a resin, each of the control terminals includes: in a partial portion of the control terminal, an exposed face exposed from the holding member; a contact face that is located opposite to the exposed face and is in contact with the holding member; a first side face located between the exposed face and the contact face; and a second side face located between the exposed face and the contact face, and a first space is formed lateral to the first side face, and a second space is formed lateral to the second side face.

(13) In the circuit assembly according to the present embodiment, the control terminals serve as insert articles that form one piece with the holding member that is made of a resin, and when the holding member is molded, a molten resin fills up the vicinity of the control terminals. The contact face of part of a control terminal comes into contact with the holding member, and the part of the control terminal forms one piece with the holding member. The exposed face of part of the control terminal is formed by coming into contact with a molding die.

(14) When the die is provided with projections on both sides of part of the control terminal, after the molding, the first space is formed lateral to the first side face, and the second space is formed lateral to the second side face. As a result of the die including projections on both sides of part of the control terminal in this way, a molten resin is unlikely to enter the space between the portion to serve as the exposed face and the die. Therefore, during molding of the holding member, it is possible to suppress such a situation that due to the injection pressure of a molten resin, the control terminals move or deform.

(15) Second Aspect

(16) Preferably, the holding member includes: a thick portion that is in contact with the contact face; a first thin-film portion that is made of a resin and covers the first side face; and a second thin-film portion that is made of a resin and covers the second side face, a side face of the first thin-film portion is exposed in the first space, and a side face of the second thin-film portion is exposed in the second space. With this configuration, not only the contact face of the control terminal, but also the first side face and the second side face thereof are joined to the resin holding member, and thus the control terminal is unlikely to be removed from the holding member.

(17) For the molding of the holding member, the die is provided with the pair of projections, and part of the control terminal is interposed between the pair of projections. Manufacturing errors are conceivable in a first size between the pair of projections, and a second size between the first side face and the second side face of the control terminal. To place part of the control terminal between the pair of projections, the first size is preferably set to be slightly greater than the second size. As a result, small gaps are formed between the projections and the control terminal, and there is a possibility of a molten resin flowing into the small gaps. However, the small gaps are thin, and the molten resin is unlikely to enter the small gaps. Therefore, the molten resin is unlikely to enter

the space between the die and the control terminal via the small gaps. Note that when the molten resin that could enter the small gaps is cured, the cured portion will serve as the first thin-film portion and the second thin-film portion.

(18) Third Aspect

(19) Preferably, the circuit assembly further includes a control board configured to output signals to the electronic components via the control terminals, wherein each of the control terminal includes: a first terminal body portion connected to the control board; a pad portion that is electrically connected to the first terminal body portion, is provided along the holding member, and includes the exposed face and the contact face, the pad portion serving as the partial portion; and a second terminal body portion electrically connected to the pad portion and the corresponding electronic component.

(20) With this configuration, by bringing a probe of a tester for electrical testing into contact with the pad portions before the control board is connected to the first terminal body portions of the control terminals, electrical testing is conducted for each of the plurality of electronic components. Since the face (contact face) of the pad portion opposite to the exposed face is in contact with the holding member, the pad portion is prevented from deforming even if the probe of the tester is brought into contact with the exposed face.

(21) Fourth Aspect

(22) Preferably, the pad portion extends from a base portion of the first terminal body portion, and the holding member includes a first embedding portion in which a boundary region between the first terminal body portion and the pad portion of the control terminal is embedded. In this case, the embedding portion can prevent the first terminal body portion from floating upward from the holding member.

(23) Fifth Aspect

(24) Preferably, the pad portion extends from a base portion of the second terminal body portion, the holding member includes a second embedding portion in which a boundary region between the pad portion and the second terminal body portion is embedded, and the second embedding portion includes a resin face that is coplanar with the exposed face of the pad portion. In this case, since the exposed face of the pad portion and the resin face of the second embedding portion are coplanar, the molding using the die is easy in the portion of the holding member that covers the portion between the pad portion and the second terminal body portion. Also, the configuration of the die is simplified.

(25) Sixth Aspect

(26) Preferably, the second terminal body portion includes: a terminal face that is exposed from the holding member and is connected to the corresponding electronic component; and a second exposed face that is located opposite to the terminal face and is exposed from the holding member.

(27) When the holding member is made of a resin and is molded by injection molding using the die, it is sufficient to partially bring a metal member to serve as the second terminal body portion into contact with the die, and position the metal member. When the die is brought into contact with that portion from both sides so as to sandwich that portion, the terminal face and the second exposed face are formed. By bringing the die into contact with the second terminal body portion from both sides, the positioning accuracy of the second terminal body portion, particularly, with respect to the terminal face is improved. As a result, a connection failure between the terminal face and the electronic component is suppressed.

(28) Seventh Aspect

(29) An electrical junction box according to the present embodiment includes: the circuit assembly according to any one of the first to sixth aspects; and a cover that covers the circuit assembly.

(30) According to the electrical junction box of the present embodiment, during molding of the resin holding member, a movement or deformation of the control terminals due to an injection pressure of a molten resin can be suppressed.

(31) The following will describe details of the embodiment of the present disclosure with reference to the drawings. Note that at least some features of the embodiment described below may be combined with each other as appropriate.

(32) Regarding Electrical Junction Box **10**

(33) FIG. **1** is a perspective view illustrating an electrical junction box **10** including a circuit assembly **11** according to the present embodiment. The electrical junction box **10** can be installed in various devices, and in the present embodiment, the electrical junction box **10** is installed in an automobile. More specifically, the electrical junction box **10** is provided on a not-shown wiring route connecting a first on-board device and a second on-board device. The electrical junction box **10** includes the circuit assembly **11**, and a cover **12** that covers the circuit assembly **11**. The cover **12** is made of metal (aluminium or an aluminium alloy), and has a heat dissipation function of dissipating heat generated in the circuit assembly **11**. The electrical junction box **10** may further include another unit (not shown) attached to the circuit assembly **11** on a side opposite to the cover **12**.

(34) Regarding Circuit Assembly **11**

(35) FIG. **2** is a perspective view of the circuit assembly **11** when viewed from the side opposite to that in FIG. **1**. FIG. **2** shows a state in which the cover **12** is detached. In FIGS. **1** and **2**, the circuit assembly **11** includes a plurality of electronic components **15**, a plurality of control terminals **16**, one holding member **14**, and one control board **13**. FIG. **3** is an exploded view of the circuit assembly **11** from which a control board **13** is detached. The circuit assembly **11** further includes a plurality of conductive plates **17**. The plurality of conductive plates **17** and the plurality of control terminals **16** constitute a power circuit of the circuit assembly **11**.

(36) In the present embodiment, the circuit assembly **11** includes three conductive plates **17** (see FIG. **1**), and eight control terminals **16** (see FIG. **3**). The number of the control terminals **16** and the number of the electronic components **15** are the same, and one control terminal **16** is connected to one electronic component **15** in a one-to-one relationship. Note that suitable numbers of conductive plates **17**, control terminals **16**, electronic components **15**, and control boards **13** are provided, and the numbers may be changed as needed.

(37) The electronic components **15** are, for example, semiconductor relays such as field effect transistors (FETs). The following description relates to a case where the electronic components **15** are field-effect transistors, which will be referred to as “FETs”. The FETs **15** include a plurality of terminals. As shown in the enlarged view of FIG. **1**, the FET **15** includes, as a plurality of terminals, a source terminal (first terminal) **15a**, a gate terminal (second terminal) **15b**, and a drain terminal (third terminal) **15c**. The terminals are mechanically and electrically connected to the conductive plates **17** or the control terminals **16** using solder or the like.

(38) Each conductive plate **17** is manufactured by pressing a metal plate into a predetermined shape. The conductive plate **17** is a piece of metal plate, and is referred to also as a busbar. The conductive plate **17** is preferably a member made of copper such as pure copper or a copper alloy, for example. The entirety of the conductive plate **17** is a conductor, and the conductive plate **17** does not have any wiring pattern as formed on a typical printed circuit board. As shown in FIG. **1**, the circuit assembly **11** includes a first conductive plate **17-1**, a second conductive plate **17-2**, and a third conductive plate **17-3**.

(39) The control terminals **16** are manufactured by pressing metal wire materials into a predetermined shape. The control terminals **16** are terminals made of metal. The control terminals **16** are preferably members made of copper such as pure copper or a copper alloy, for example. FIG. **4** is a cross-sectional view illustrating a control terminal **16** and part of the holding member **14**. The control terminal **16** has a bent shape, and includes a first terminal body portion **31**, a pad portion **33**, and a second terminal body portion **32**. The components of the control terminal **16** will be described in detail later.

(40) In FIG. **2**, the control board **13** is an insulated substrate having a substantially rectangular

shape, and includes, on a first surface **13a** thereof, at least one control circuit unit **27**. The control circuit unit **27** has a function of outputting signals to the FETs **15** via the control terminals **16**. The FETs **15** operate based on the signals. The control circuit unit **27** has an integrated circuit including switching elements, serving as second electronic components for operating the FETs **15**, diodes, and the like. A wiring pattern **29** for electrically connecting the second electronic components to the control terminals **16** is included in the control circuit unit **27**.

(41) The control board **13** has through holes **18** through which the control terminals **16** are partially passed (see FIG. 3). By filling the through holes **18** with an electrically-conducting material, the control circuit unit **27** is electrically and mechanically connected to part of the first terminal body portions **31** of the control terminals **16**. Before the control board **13** is connected to the control terminals **16** (see FIG. 3), the plurality of control terminals **16** are not electrically connected to each other (in a non-conducting state).

(42) The holding member **14** is made of a thermoplastic resin, and is molded by injection molding. In the present embodiment, as will be described later, the holding member **14** is manufactured by insert molding that is performed with the control terminals **16** and the conductive plates **17** placed in a die **60** for injection molding (see FIG. 7). Therefore, the control terminals **16** and the conductive plates **17** are partially embedded in the holding member **14**, and the control terminals **16**, the conductive plates **17**, and the holding member **14** are formed as one piece. The control terminals **16** and the conductive plates **17** are insert articles that forms one piece with the resin holding member **14**.

(43) The holding member **14** is made of, for example, polyphenylene sulfide (PPS), polybutylene terephthalate (PBT), nylon, polypropylene (PP), polyethylene (PE), or the like, and has insulating properties. The holding member **14** holds the eight control terminals **16** and the three conductive plates **17**, and functions as a case of the circuit assembly **11**. As shown in FIG. 3, the holding member **14** has a housing space **30** that is open toward the control board **13**. The first terminal body portions **31** and the pad portions **33** of the control terminals **16** are provided in the housing space **30**. The control board **13** is attached to the first terminal body portions **31** so as to close the housing space **30**.

(44) The holding member **14** includes a plate-shaped base portion **19** provided in the center of the holding member **14**, and a frame portion **20** provided surrounding of the base portion **19**. As shown in FIG. 1, main portions **22** of the conductive plates **17** on which the FETs **15** are mounted are provided on the base portion **19**. The main portions **22** of the three conductive plates **17** are arranged side by side in a flat manner on the base portion **19** without overlapping each other. Each of the three conductive plates **17** are provided at separate positions, and are not in contact with each other. The three conductive plates **17** and the control terminals **16** are provided at separate positions, and are not in contact with each other.

(45) In the present embodiment (see FIG. 1), the source terminals **15a** of the FETs **15** are connected to the third conductive plate **17-3** located in the center. The drain terminals **15c** of the FETs **15** is connected to the first conductive plate **17-1** or the second conductive plate **17-2**. As shown in the enlarged view of FIG. 1, the third conductive plate **17-3** located in the center has a hole **28** penetrating therethrough, and part of the second terminal body portion **32** of the control terminal **16** is exposed from the hole **28**. The gate terminal **15b** of the FET **15** is connected to the part of the second terminal body portion **32** exposed from the hole **28**. With this, the gate terminal **15b** of the FET **15** and the control terminal **16** are electrically connected to each other.

(46) The frame portion **20** has fixation holes **39** used to fix the cover **12** thereto. An end portion **21A** of the first conductive plate **17-1** and an end portion **21B** of the second conductive plate **17-2** protrude outward from the frame portion **20**. Not-shown cables or the like is connected to the end portions **21A** and **21B**.

(47) Control Terminal **16**

(48) FIG. 5 is a perspective view illustrating control terminals **16** and part of the holding member

14. The control terminals **16** have a bent shape as described above, and each include the first terminal body portion **31**, the pad portion **33**, and the second terminal body portion **32**. In the present embodiment, the first terminal body portion **31**, the pad portion **33**, and the second terminal body portion **32** are formed by bending and molding a single metal member. That is to say, the first terminal body portion **31** is constituted by a part of an elongated plate-shaped metal member, the pad portion **33** is constituted by another part of the metal member, and the second terminal body portion **32** is constituted by yet another part of the metal member that is different from the parts constituting the first terminal body portion **31** and the pad portion **33**.

(49) Each of the first terminal body portion **31**, the pad portion **33**, and the second terminal body portion **32** are rectangular in a traverse cross section (see FIG. 6). FIG. 6 is a cross-sectional view taken along a line V in FIG. 5. As shown in FIG. 5, the first terminal body portion **31** is linear. The control terminal **16** is bent at a right angle between the first terminal body portion **31** and the pad portion **33**. The pad portion **33** extends from a base portion **31a** of the first terminal body portion **31**, and is linear. The control terminal **16** is bent at a right angle between the second terminal body portion **32** and the pad portion **33**. The pad portion **33** extends from a base portion **32a** of the second terminal body portion **32**, and is linear. The second terminal body portion **32** is U-shaped.

(50) First Terminal Body Portion **31**

(51) The first terminal body portions **31** are connected to the control board **13** (see FIGS. 2 and 3). A leading ends of the first terminal body portions **31** are inserted into the through holes **18** formed in the control board **13**, and are electrically connected to the wiring pattern **29** provided on the control board **13**. As described above, the control circuit units **27** included in the control board **13** output signals to the FETs **15** via the control terminals **16**. The signals are output from the control board **13** to the first terminal body portions **31**. The signals are given to the gate terminals **15b** of the FETs **15** from the first terminal body portions **31** via the pad portions **33** and the second terminal body portions **32** (see FIG. 5).

(52) The first terminal body portions **31** have the shape of an elongated plate. Each first terminal body portion **31** protrudes from a corresponding first embedding portion **23**, which is a part of the holding member **14**. A portion of the first terminal body portion **31** that protrudes from the first embedding portion **23** is entirely exposed. That is to say, as shown in FIG. 5, the first terminal body portion **31** includes a first face **41** that is exposed from the holding member **14** on one side in the thickness direction of the first terminal body portion **31**, and a second face **42** that is located opposite to the first face **41** and is exposed from the holding member **14**. Furthermore, two side faces **43** of the first terminal body portion **31** that are located between the first face **41** and the second face **42** are also exposed. The first terminal body portion **31** extends from the first embedding portion **23** in a cantilever fashion before the control board **13** is connected to the first terminal body portion **31**.

(53) Pad Portion **33**

(54) The pad portion **33** extends from the base portion **31a** of the first terminal body portion **31**, and the pad portion **33** serves as one piece with and is electrically connected to the first terminal body portion **31**. The pad portion **33** has the shape of an elongated plate. As shown in FIG. 5, the pad portion **33** is provided along the holding member **14**, and is exposed from part of the holding member **14**. The pad portion **33** is provided along a thick portion **50** included in the holding member **14**.

(55) As shown in FIGS. 4 and 5, the pad portion **33** includes a first exposed face **34** exposed from the holding member **14**, and a contact face **35** located opposite to the first exposed face **34**. The first exposed face **34** has an elongated shape. Specifically, for example, the first exposed face **34** has an elongated rectangular shape, and includes a first side **34a**, and a second side **34b** that is more than twice as long as the first side **34a**. The contact face **35** is entirely in contact with the holding member **14**, and is a non-exposed face that is not exposed from the holding member **14**. As described above, the holding member **14** is manufactured by insert molding that is performed with

the control terminals **16** placed in the die **60** for injection molding (see FIG. 7). Therefore, the pad portion **33** is joined to part of the holding member **14** on the contact face **35**.

(56) As shown in FIG. 6, the pad portion **33** includes, in addition to the first exposed face **34** and the contact face **35**, a first side face **36** and a second side face **37**. The first side face **36** is a face located between the first exposed face **34** and the contact face **35**. The second side face **37** is a face that is located opposite to the first side face **36** and is located between the first exposed face **34** and the contact face **35**.

(57) In the present embodiment, the first side face **36** is covered by a first thin-film portion **51**, which is made of a resin, and the second side face **37** is covered by a second thin-film portion **52**, which is made of a resin. The first thin-film portion **51** and the second thin-film portion **52** constitute part of the holding member **14**. The first thin-film portion **51** and the second thin-film portion **52** are both film-shaped resin portions, and are thinner than the thick portion **50** to which the contact face **35** is adjoined. A first space K1 is formed lateral to the first side face **36** covered by the first thin-film portion **51**. A second space K2 is formed lateral to the second side face **37** covered by the second thin-film portion **52**.

(58) A side face **51a** of the first thin-film portion **51** is exposed in the first space K1. A side face **52a** of the second thin-film portion **52** is exposed in the second space K2. The holding member **14** includes a first bulge portion **57** bulging from the thick portion **50**. The first bulge portion **57** and the pad portion **33** are provided at separate positions. The space formed between the first bulge portion **57** and the pad portion **33** (a first thin portion **51**) serves as the first space K1. The holding member **14** includes a second bulge portion **58** bulging from the thick portion **50**. The second bulge portion **58** and the pad portion **33** are provided at separate positions. The space formed between the second bulge portion **58** and the pad portion **33** (a second thin portion **52**) serves as the second space K2.

(59) Second Terminal Body Portion **32**

(60) As shown in FIG. 5, the second terminal body portion **32** is provided so as to be continuous with the pad portion **33**. The second terminal body portion **32** has a shape (U-shape) such that an elongated plate is bent. The base portion **32a** of the second terminal body portion **32** is continuous with the pad portion **33**, and the second terminal body portion **32** is electrically connected to the pad portion **33**. The second terminal body portion **32** includes a first area **46** continuous with the pad portion **33**, a second area **47** provided at a distance to and in parallel to the first area **46**, and an intermediate area **48** that connects the first area **46** and the second area **47**. The second area **47** is exposed from the holding member **14**, and the intermediate area **48** has a face exposed from the holding member **14**.

(61) The gate terminal **15b** of the FET **15** is connected to the intermediate area **48** using solder or the like. With this, the second terminal body portion **32** is mechanically and electrically connected to the FET **15**. The second terminal body portion **32** includes, on the intermediate area **48** thereof, a terminal face **45** exposed from the holding member **14**, and a second exposed face **44** that is located opposite to the terminal face **45** and is exposed from the holding member **14**. The FET **15** (gate terminal **15b**) is connected to the terminal face **45**.

(62) Holding Member **14**

(63) FIG. 7 is a cross-sectional view illustrating part of the die **60** for molding the resin holding member **14**. With the portion of the die **60** that is shown in FIG. 7, the cross-sectional structure of the circuit assembly **11** shown in FIG. 6 is obtained. In the circuit assembly **11** of the present embodiment, the control terminals **16** and the conductive plates **17** function as insert articles that forms one piece with the resin holding member **14**. When molding the holding member **14**, a cavity **60a** formed inside the die **60** is filled with a molten resin. The molten resin also fills up the vicinity of the control terminals **16**. With this molding, the contact face **35** of the pad portion **33** included in the control terminal **16** is in contact with the holding member **14** (see FIG. 6), and the pad portion **33** is formed as one piece with the holding member **14**. In the cavity **60a**, the molten resin also fills

up the vicinity of the conductive plates **17**. With this, the conductive plates **17** are in contact with the holding member **14** (see FIG. **6**), and is formed as one piece with the holding member **14**.

(64) As shown in FIG. **7**, the exposed face **34** of the pad portion **33** is in contact with the die **60** during the molding, and is completed when separating the metal member from the die **60** at the time of removal of the die **60**. The die **60** is provided with projections **61** on both sides of the pad portion **33**. With these projections **61**, after the molding (see FIG. **6**), the first space K1 is formed lateral to the first side face **36** of the pad portion **33**, and the second space K2 is formed lateral to the second side face **37**. The projections **61** are portions constituting the die **60**.

(65) The cross-sectional shape of the first space K1 and the second space K2 matches the cross-sectional shape of the projections **61**. Accordingly, since the die **60** is provided with the pad portions **61** on both sides of the pad portion **33**, a molten resin is unlikely to enter the space between the portion to serve as the exposed face **34** and the die **60**. That is to say, the molten resin flows through and fills up the cavity **60a**, and at that time, the projections **61** function as barriers for hiding the pad portion **33**, making the flow of the resin unlikely to affect the pad portion **33**.

(66) Therefore, during molding of the holding member **14**, it is possible to suppress such a situation that due to the injection pressure of a molten resin, the control terminals **16** including the pad portion **33** move in the cavity **60a** or deform. As a result, it is possible to prevent the control terminals **16** (pad portion **33**) and the conductive plates **17** from coming into contact with each other (shorting).

(67) As shown in FIG. **6**, after molding, the holding member **14** includes a resin thick portion **50** that is in contact with the contact faces **35** of the pad portions **33**, a resin first thin-film portion **51** that covers the first side faces **36**, and a resin second thin-film portion **52** that covers the second side faces **37**. The first thin-film portion **51** and the second thin-film portion **52** are sufficiently thinner than the thick portion **50**. The side face **51a** of the first thin-film portion **51** is exposed in the first space K1. The side face **52a** of the second thin-film portion **52** is exposed in the second space K2. The pad portion **33** has a configuration in which in addition to the contact face **35**, the first side face **36** and the second side face **37** are joined to the resin holding member **14**. As a result, the control terminals **16** including the pad portion **33** are unlikely to be removed from the holding member **14**.

(68) As described above (see FIG. **7**), for molding the holding member **14**, the die **60** is provided with the pair of projections **61**. A portion (pad portion **33**) of the control terminal **16** is interposed between these projections **61**. A first size L1 between the pair of projections **61**, and a second size L2 between the first side face **36** and the second side face **37** of the control terminal **16** may have a manufacturing error. To interpose the pad portion **33** between the pair of projections **61**, the first size L1 is preferably set to be slightly greater than the second size L2. As a result, small gaps e are formed between the projections **61** and the pad portion **33**. A molten resin may flow into the small gaps e.

(69) However, the small gaps e are thin, and the molten resin is unlikely to enter the small gaps e. Therefore, the molten resin is unlikely to enter the space between the die **60** and the first exposed face **34** of the pad portion **33** via the small gaps e. Note that when the molten resin that could enter the small gaps e is cured, the cured portion will serve as the first thin-film portion **51** and the second thin-film portion **52**.

(70) The gap size of the small gaps e is preferably set to a value such that a molten resin is unlikely to enter the small gaps e. With this, corners (edges) **38** of the pad portion **33** that are formed between the first exposed face **34** and the first side face **36** (or the second side face **37**) are exposed. The corners **38** are preferably not covered by the first thin-film portion **51** or the second thin-film portion **52**. The first thin-film portion **51** and the second thin-film portion **52** are lower than the first exposed face **34** using the thick portion **50** as a standard.

(71) As shown in FIGS. **4** and **5**, the first bent portion **25** located between the first terminal body portion **31** and the pad portion **33** is embedded in the first embedding portion **23**, which is made of

a resin. The first embedding portion **23** is part of the holding member **14**. That is to say, the holding member **14** includes the first embedding portion **23** in which a first boundary region J1 of the control terminal **16** located between the first terminal body portion **31** and the pad portion **33** is embedded. Part of the pad portion **33** and most part of the first terminal body portion **31** are exposed from the holding member **14**, but the first embedding portion **23** can prevent the first terminal body portion **31** from floating upward from the holding member **14**.

(72) A second bent portion **26** located between the second terminal body portion **32** and the pad portion **33** is embedded in a second embedding portion **24**, which is made of a resin. The second embedding portion **24** is another part of the holding member **14**. That is to say, the holding member **14** includes the second embedding portion **24** in which a second boundary region J2 of the control terminal **16** located between the pad portion **33** and the second terminal body portion **32** is embedded. The second embedding portion **24** can prevent the second terminal body portion **32** and the pad portion **33** from floating upward from the holding member **14**.

(73) The first space K1 and the second space K2 are formed via the first thin-film portion **51** and the second thin-film portion **52** on both sides of the pad portion **33** (see FIG. 6) in a direction orthogonal to the longitudinal direction of the pad portion **33**. That is to say, the holding member **14** is cut off on both sides of the pad portion **33**. In contrast, as shown in FIG. 5, the first embedding portion **23** is provided on the first terminal body portion **31** side of the pad portion **33** in the longitudinal direction of the pad portion **33**, and resin embedding is made. The second embedding portion **24** is provided on the second terminal body portion **32** side of the pad portion **33** in the longitudinal direction of the pad portion **33**, and resin embedding is made.

(74) The second bent portion **26** between the pad portion **33** and the second terminal body portion **32** is embedded in the second embedding portion **24**. The second embedding portion **24** includes a resin face **54** that is coplanar (constitutes the same plane) with the first exposed face **34** of the pad portion **33**. Since the first exposed face **34** of the pad portion **33** and the resin face **54** of the second embedding portion **24** are coplanar, the molding using the die **60** (see FIG. 8) is easy in the portion of the holding member **14** that covers the portion (base portion **32a**) between the pad portion **33** and the second terminal body portion **32**. Also, the configuration of the die **60** is simplified.

(75) Two faces of the intermediate area **48**, which is a midway portion of the second terminal body portion **32**, are exposed from the holding member **14**. That is to say, the second terminal body portion **32** includes, on the intermediate area **48**, a terminal face **45** that is exposed from the holding member **14** and is connected to the FET **15**, and a second exposed face **44** that is located opposite to the terminal face **45** and is exposed from the holding member **14**.

(76) FIG. 8 is a cross-sectional view illustrating an aspect in which the holding member **14** is molded with the control terminal **16** serving as an insert article. FIG. 8 shows the second terminal body portion **32** of the control terminal **16** and the vicinity thereof, together with the die **60**. As described above, the holding member **14** is made of a resin, and is molded by injection molding using the die **60**. A metal member to serve as the second terminal body portion **32** needs only to be partially brought into contact with the die **60** and be positioned. Here, as shown in FIG. 8, the die **60** is brought into contact with a partial portion **32b** of the second terminal body portion **32** so that the partial portion **32b** is sandwiched by the die **60** from two sides. The terminal face **45** and the second exposed face **44** are formed on the portions of the second terminal body portion **32** with which the die **60** is in contact.

(77) By bringing a first portion **60b** and a second portion **60c** of the die **60** into contact with the partial portion **32b** of the second terminal body portion **32** from both sides in this way, the positioning accuracy of the second terminal body portion **32**, particularly the terminal face **45**, is improved. As a result, a connection failure between the terminal face **45** and the FET **15** (gate terminal **15b**) is suppressed.

(78) Regarding Electrical Testing of Electronic Components (FET) **15**

(79) In the process for manufacturing the circuit assembly **11**, electrical testing is conducted to

determine whether or not there is any connection failure in the FETs **15**. In the case of the circuit assembly **11** of the present embodiment, as shown in FIG. **3**, the pad portions **33** included in the control terminals **16** are housed in the housing space **30** of the holding member **14**, but the pad portions **33** are exposed to the outside before the control board **13** is connected to the control terminals **16**. Therefore, before connecting the control board **13** to the control terminals **16**, by bringing a probe of a not-shown tester into contact with the pad portions **33**, electrical testing is conducted for each of the plurality of FETs **15**.

(80) For this reason, each of the control terminals **16** includes the pad portion **33**, and the second terminal body portion **32** that is electrically connected to the pad portion **33** and the corresponding FET **15**. The pad portions **33** are provided along the resin holding member **14**, and are exposed from part of the holding member **14**. By bringing the probe of the tester into contact with the pad portions **33**, the electrical testing can be conducted for the plurality of FETs **15**. Therefore, the control board **13** does not need to include any test circuit component **99** (see FIG. **9**) as is conventionally provided. As a result of omission of such a test circuit component **99**, it is possible to realize a cost reduction of the circuit assembly **11**, as well as down-sizing of the control board **13**.

(81) In the present embodiment, the first terminal body portion **31** (see FIG. **4**) includes the first face **41** exposed from the holding member **14**, and the second face **42** that is located opposite to the first face **41** and is exposed from the holding member **14**. The first terminal body portion **31** has the shape of a thin and elongated plate, and both the first face **41** and the second face **42** are exposed from the holding member **14**. Accordingly, if the probe of the tester is brought into contact with the first face **41** or the second face **42**, the first terminal body portion **31** may deform. When the first terminal body portion **31** has deformed, connection to the control board **13** may malfunction.

(82) In contrast, the pad portion **33** includes the first exposed face **34** exposed from the holding member **14**, and the contact face **35** that is located opposite to the first exposed face **34** and is in contact with the holding member **14**. Since the face (contact face **35**) of the pad portion **33** located opposite to the first exposed face **34** is in contact with the holding member **14**, even if the probe of the tester is brought into contact with the first exposed face **34**, the force caused by the contact is supported by the holding member **14**, preventing the pad portion **33** from deforming. It is sufficient to bring the probe of the tester into contact with a target region Q (see FIG. **5**) of the first exposed face **34** of the pad portion **33** that is away from the first terminal body portion **31** so that the probe does not interfere with the first terminal body portion **31**. The first exposed face **34** of the pad portion **33** has an elongated shape that includes not only the target region Q but also a portion including this target region Q. For example, the first exposed face **34** of the pad portion **33** has the second side **34b** that is more than three times as long as the first side **34a**.

(83) This is because the holding member **14** is made of a resin, and as described with reference to FIG. **7**, the holding member **14** is molded by injection molding using the die **60**, and by bringing part of a metal material to serve as the pad portion **33** into contact with the die **60**, the contact part serves as the first exposed face **34**. That is to say, when the metal member is brought into contact with the die **60**, the longer the contact part is, the more reliably the metal member is positioned. Accordingly, by setting the first exposed face **34** of the pad portion **33** as being elongated, the contact area of the metal member with the die **60** increases, and the metal member becomes stable, suppressing occurrence of defective products.

(84) Other Configurations

(85) In the present embodiment, with respect to each control terminal **16**, the first terminal body portion **31**, the pad portion **33**, and the second terminal body portion **32** are formed by bending and molding a single metal member. Therefore, it is not necessary to join the first terminal body portion **31**, the pad portion **33**, and the second terminal body portion **32** together. Note that one or all of the first terminal body portion **31**, the pad portion **33**, and the second terminal body portion **32** may be a separate member or separate members, and the control terminals **16** may join these portions

together using welding or the like.

(86) As shown in FIG. 4, the pad portion 33 is provided along the holding member 14, and the contact face 35 thereof is entirely in contact with the holding member 14. As another configuration, part of the contact face 35 may be exposed from the holding member 14.

(87) Although the case has been described in which the first exposed face 34 of the pad portion 33 has an elongated and rectangular shape, the first exposed face 34 may also have another shape such as a circular shape or an oval shape.

(88) In the present embodiment (see FIG. 6), the first side face 36 of the pad portion 33 is covered by the first thin-film portion 51, which is made of a resin, and the second side face 37 is covered by the second thin-film portion 52, which is made of a resin. As a modification, either or both of the first thin-film portion 51 and the second thin-film portion 52 may be omitted.

(89) The first space K1 and the second space K2 respectively formed lateral to the first side face 36 and the second side face 37 of the pad portion 33 may have a shape other than the shown shape (cross-sectional shape).

(90) The electronic components 15 (first electronic components 15) may be components other than field-effect transistors, and may be, for example, mechanical relays. Also in this case, when the circuit assembly 11 is completed, the electronic components 15 and the control board 13 are electrically connected to each other, signals are given from the control board 13 to the electronic components 15 via the control terminals 16, and the electronic components 15 operate based thereon. Note that the electrical testing are conducted before connecting the control board 13.

(91) The above-described embodiments are examples in all respects, and are non-limiting. The claimed scope of the present disclosure is indicated by not the above-described embodiments but the claims, and includes all modifications within the scope equivalent to the scope of the claims.

Claims

1. A circuit assembly comprising: a plurality of electronic components; a plurality of control terminals electrically connected respectively to the plurality of electronic components; and a holding member that has insulating properties and holds the plurality of control terminals, wherein the control terminals serve as insert articles that form one piece with the holding member that is made of a resin, each of the control terminals includes, in a partial portion of the control terminal: an exposed face exposed from the holding member; a contact face that is located opposite to the exposed face and is in contact with the holding member; a first side face located between the exposed face and the contact face; and a second side face located between the exposed face and the contact face, and a first space is formed lateral to the first side face, and a second space is formed lateral to the second side face; and wherein the holding member includes: a thick portion that is in contact with the contact face; a first thin-film portion that is made of a resin and covers the first side face; and a second thin-film portion that is made of a resin and covers the second side face, and wherein a side face of the first thin-film portion is exposed in the first space, and a side face of the second thin-film portion is exposed in the second space.

2. The circuit assembly according to claim 1, further comprising: a control board configured to output signals to the electronic components via the control terminals, wherein each of the control terminal includes: a first terminal body portion connected to the control board; a pad portion that is electrically connected to the first terminal body portion, is provided along the holding member, and includes the exposed face and the contact face, the pad portion serving as the partial portion; and a second terminal body portion electrically connected to the pad portion and the corresponding electronic component.

3. The circuit assembly according to claim 2, wherein the pad portion extends from a base portion of the first terminal body portion, and the holding member includes a first embedding portion in which a boundary region between the first terminal body portion and the pad portion of the control

terminal is embedded.

4. The circuit assembly according to claim 2, wherein the pad portion extends from a base portion of the second terminal body portion, the holding member includes a second embedding portion in which a boundary region between the pad portion and the second terminal body portion is embedded, and the second embedding portion includes a resin face that is coplanar with the exposed face of the pad portion.

5. The circuit assembly according to claim 2, wherein the second terminal body portion includes: a terminal face that is exposed from the holding member and is connected to the corresponding electronic component; and a second exposed face that is located opposite to the terminal face and is exposed from the holding member.

6. An electrical junction box comprising: the circuit assembly according to claim 1; and a cover that covers the circuit assembly.

7. The circuit assembly according to claim 3, wherein the pad portion extends from a base portion of the second terminal body portion, the holding member includes a second embedding portion in which a boundary region between the pad portion and the second terminal body portion is embedded, and the second embedding portion includes a resin face that is coplanar with the exposed face of the pad portion.

8. The circuit assembly according to claim 3, wherein the second terminal body portion includes: a terminal face that is exposed from the holding member and is connected to the corresponding electronic component; and a second exposed face that is located opposite to the terminal face and is exposed from the holding member.

9. The circuit assembly according to claim 4, wherein the second terminal body portion includes: a terminal face that is exposed from the holding member and is connected to the corresponding electronic component; and a second exposed face that is located opposite to the terminal face and is exposed from the holding member.

10. The electrical junction box according to claim 6, wherein the holding member includes: a thick portion that is in contact with the contact face; a first thin-film portion that is made of a resin and covers the first side face; and a second thin-film portion that is made of a resin and covers the second side face, a side face of the first thin-film portion is exposed in the first space, and a side face of the second thin-film portion is exposed in the second space.

11. The electrical junction box according to claim 6, further comprising: a control board configured to output signals to the electronic components via the control terminals, wherein each of the control terminal includes: a first terminal body portion connected to the control board; a pad portion that is electrically connected to the first terminal body portion, is provided along the holding member, and includes the exposed face and the contact face, the pad portion serving as the partial portion; and a second terminal body portion electrically connected to the pad portion and the corresponding electronic component.

12. The electrical junction box to claim 11, wherein the pad portion extends from a base portion of the first terminal body portion, and the holding member includes a first embedding portion in which a boundary region between the first terminal body portion and the pad portion of the control terminal is embedded.

13. The electrical junction box to claim 11, wherein the pad portion extends from a base portion of the second terminal body portion, the holding member includes a second embedding portion in which a boundary region between the pad portion and the second terminal body portion is embedded, and the second embedding portion includes a resin face that is coplanar with the exposed face of the pad portion.

14. The electrical junction box according to claim 11, wherein the second terminal body portion includes: a terminal face that is exposed from the holding member and is connected to the corresponding electronic component; and a second exposed face that is located opposite to the terminal face and is exposed from the holding member.

